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Projektbericht

PAPER TITLE

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Abstract

This is a nice abstract.

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List of Abbreviations

IP Integer Programming

KP Knapsack Problem

TSP Traveling Salesman Problem

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1 Introduction

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

The main contributions of this work are the following. On the one hand, it serves as a suitable L^AT_EX template (with reasonable formatting options) that can be used for academic papers. On the other hand, it provides a loose guideline on how one might arrange and structure such a work along with some concrete examples, e.g., mathematical formulations and figures.

The remainder of this paper is organized as follows. We begin in Section 2 with a short introduction to the Knapsack Problem. In more detail, we discuss a mathematical formulation, the computational complexity, and dynamic programming approaches. Afterwards, in Section 3, we dedicate ourselves to the Traveling Salesman Problem. Finally, concluding remarks will be drawn in Section 4.

2 The Knapsack Problem

In the Knapsack Problem (KP) (refer to Martello & Toth, 1990), we are given a set of items $\mathcal{I} = \{1, \dots, n\}$. Each item $i \in \mathcal{I}$ has a positive weight $w_i \in \mathbb{R}^+$ and value $v_i \in \mathbb{R}^+$. Moreover, it is assumed that the capacity of our knapsack is $C \in \mathbb{R}^+$. In the KP, the objective is to pack items into our knapsack such that the cumulative value is maximized while respecting the capacity restriction.

2.1 Integer Programming Model

In order to model the problem, we introduce a binary variable $x_i \forall i \in \mathcal{I}$ that indicates if item i is packed into our knapsack. Using these variables, the KP can be modeled through the following Integer Programming (IP) model:

$$\max \quad \sum_{i=1}^n v_i x_i, \quad (1)$$

$$\text{subject to} \quad \sum_{i=1}^n w_i x_i \leq C, \quad (2)$$

$$x_i \in \{0, 1\} \quad \forall i \in \mathcal{I}. \quad (3)$$

In this model, the objective is given by (1) and asks that the total value of the selected items is maximized. Constraint (2) guarantees that the capacity is respected. Finally, constraints (3) restrict the domain of the variables to binary.

2.2 Computational Complexity

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2.3 Dynamic Programming Approaches

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Algorithm 1: Example code

```
/* This is a comment.  This type of comment can span multiple lines.  
   However, comments should only be used sparsely.                               */  
1 for  $i$  in range (1,  $n$ ) do  
2   | Do something                               // This is another comment.
```

Algorithmus 1 ist so toll!

3 The Traveling Salesman Problem

Assume that we are given a complete graph $\mathcal{G}(V, E)$ where V is the set of vertices and E is the set of edges. For each pair of vertices $i, j \in V, i \neq j$ the edge $(i, j) \in E$ is associated with a distance $d_{ij} \in \mathbb{R}^+$. In the Traveling Salesman Problem (TSP), we are tasked with finding a *tour* with minimal distance that visits every vertex exactly once and then returns to the point of origin, i.e., a *hamiltonian cycle* with minimal weight (see Figure 1).

In fact, the TSP is a well-known combinatorial optimization problem that has already been studied for a long time (see, e.g., Dantzig et al., 1954; Lin, 1965). Dantzig et al. (1954) introduced a lot of stuff! Bla bla bla here is another parenthetical citation (Lin, 1965).

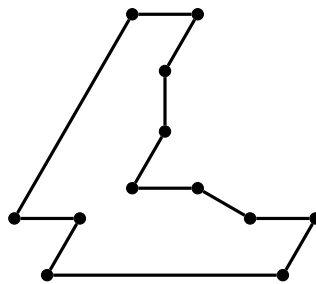


Figure 1: A hamiltonian cycle on a graph $\mathcal{G}(V, E)$ that consists of $|V| = 12$ vertices.

4 Conclusion

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Eidesstattliche Erklärung

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